The Budget Governor Principle

Technical Note on Control-Knob Sufficiency, Low-Dimensional Generalization, Inverse Recovery, and Symmetry-Limited Pose-Free Inference

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Abstract

The Budget Governor Principle (BGP) is the experimentally established control-parameter law for the symmetric constant-sum two-source Euclidean process. In that base case, BGP is the latent control parameter `e = c/a` for normalized geometry and `b/a = sqrt(1-e^2)` is the corresponding transverse residue. Controlled computational studies in this repository show that this parameter governs normalized shape, remains recoverable from noisy boundary observations when the source positions are known, and outperforms raw separation and raw budget variables under scale shift. Additional studies show that the same budget logic extends in a structured way: asymmetry upgrades the family from one knob to two, the hyperbola case yields a deficit-side twin, controlled anisotropy adds a medium parameter removable by whitening, and equal-weight and weighted three-source families are governed by compact normalized control objects. In weighted multi-source and anisotropic settings those control objects form operational latent variables recoverable from boundary data. The main current solver challenge is the pose-free anisotropic inverse, where geometry and weights remain comparatively stable but anisotropy becomes weakly identified under hidden rotation. Matched ambiguity studies, oracle alignment, and alignment failure maps localize that solver challenge to symmetry handling rather than missing signal.

Keywords: ellipse eccentricity; conic geometry; inverse problems; latent variables; anisotropy; multi-source geometry; scale collapse; shape analysis

1 Opening Statement

BGP is the latent control parameter `e = c/a` for the symmetric constant-sum two-source Euclidean process.

The repository establishes that base case, extends the same budget logic into compact low-dimensional control objects for richer families, and demonstrates boundary-only recovery of those control objects in weighted multi-source settings (Lopez III 2026).

The established result set is:

- in the symmetric two-source base case, `e = c/a` governs normalized geometry
- in that same setting, boundary data recovers `e` and `e` outperforms raw separation and raw budget variables under scale shift
- beyond the base case, asymmetry, anisotropy, and multi-source families are still governed by compact normalized control objects

- the main open solver challenge is pose-free anisotropic inversion, where hidden rotation selectively weakens anisotropy recovery

2 Core Proposal

The narrowest version of BGP is the symmetric constant-sum two-source Euclidean process:

$$\|x - F_1\| + \|x - F_2\| = 2a$$

with focal half-separation `c`. In that setting, the governor variable is

$$e = \frac{c}{a},$$

defined as a normalized separation load. The corresponding transverse residue is

$$\frac{b}{a} = \sqrt{1 - e^2}.$$

The core BGP reading is therefore:

> normalized source separation relative to total budget governs how much geometric freedom remains after structural separation cost is paid.

In the symmetric ellipse case, that control object collapses to one scalar. In richer settings, the scalar does not survive unchanged, but the underlying budget logic survives in the form of compact control manifolds built from normalized placement, participation, and medium structure.

3 Why This Matters Computationally

The computational importance of BGP is narrower and stronger than static shape description:

1. the symmetric ratio `e = c/a` is the sufficient organizing variable for normalized geometry in the tested base case,
2. that variable is operational in recovery and prediction,
3. once symmetry is broken or the source family expands, the collapse does not dissolve chaotically but upgrades to low-dimensional control objects,
4. and in weighted multi-source settings those compact objects can be inferred from boundary data.

The project establishes that budget-normalized latent structure governs the tested shape families in a reusable way.

4 Mathematical Core: The Symmetric Control Knob

The foundation experiment in this repository tests the strongest mathematical heart of the idea:

- process reconstruction from the constant-sum two-circle construction,
- scale collapse at fixed `e`,

- one-dimensional response curves for normalized observables,
- and a full separation-budget phase map.

The result is clean. Across the sweep:

- maximum ellipse-equation residual was `1.5876e-14`,
- maximum pairwise scale-collapse error after normalization was `3.9736e-08`,
- and normalized width, perimeter, and tip-response observables behaved as functions of `e` alone within numerical tolerance.

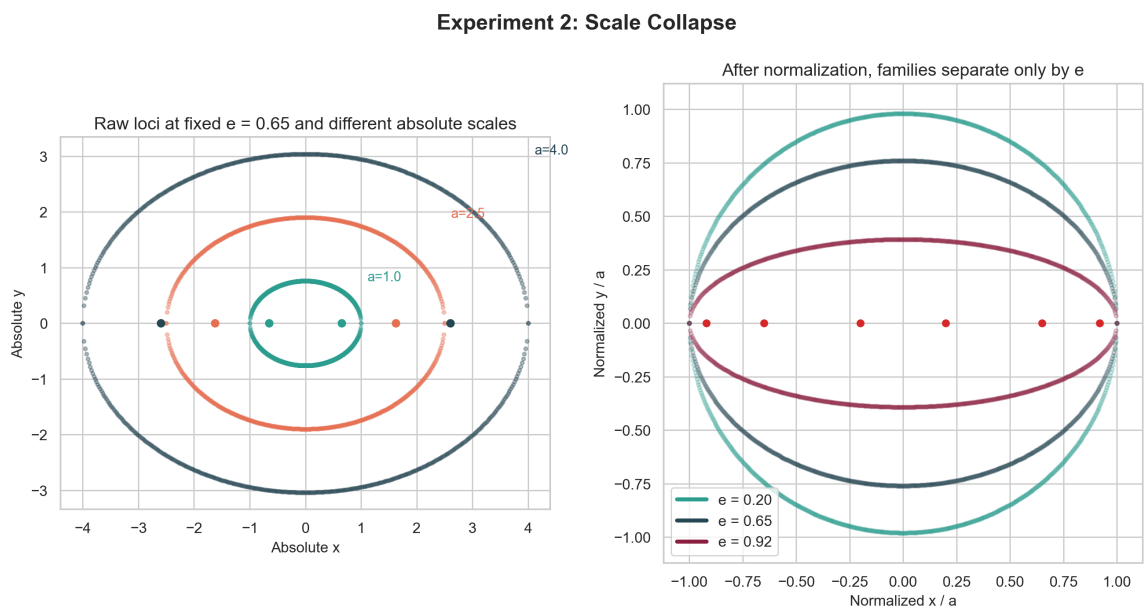

Figure 1: Scale collapse in the symmetric two-source case. Fixed `e` produces the same normalized locus across absolute scales, which is the cleanest visual statement of one-knob sufficiency in the tested process.

This is the narrowest claim the repository now establishes:

> under the symmetric constant-sum two-source Euclidean process, `e = c/a` is the sufficient control variable for normalized geometry.

That is the base-case result.

5 Operational Evidence in the Symmetric Setting

The next question is whether the control knob is operational, not only compact.

The known-source inverse experiment shows that `e` is recoverable from noisy, partial, and sparse boundary observations with high accuracy. Mean absolute recovery error ranged from `1.46e-4` in the easiest setting to `3.46e-3` in the harshest tested setting, and even the worst 95th-percentile error remained about `1.26e-2`.

The same study also compared `e` against raw alternatives under a scale-held-out prediction split. For normalized perimeter, for example, test RMSE was `1.90e-4` using `e` alone, versus `5.59e-1` for a low-capacity model on `(d, S)`, `3.64` for raw `d`, and `5.65` for raw `S`.

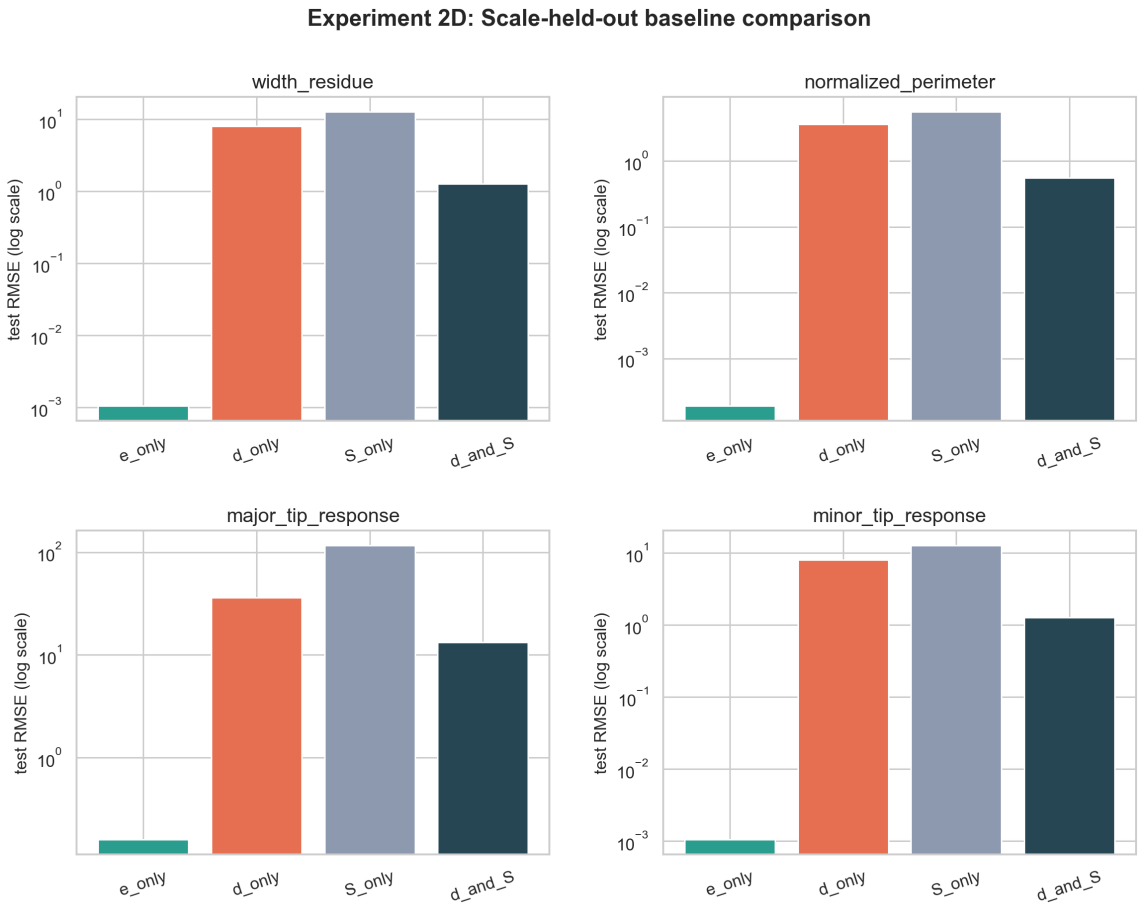

Figure 2: Scale-held-out baseline comparison. The normalized ratio e preserves predictive power across scale far better than raw separation, raw budget, or a low-capacity model on both together.

This is the first operational inverse result in the repository.

The edge-regime study sharpened that result further. Near `e = 0`, several shape summaries are first-order flat, so low-depletion systems are intrinsically hard to distinguish from shape alone. Near `e = 1`, width and major-tip response become much sharper probes than perimeter. The measured crossover points were:

- `e = 0.57735` for major-tip response,
- `e = 0.70711` for width,
- `e = 0.90891` for perimeter.

So in the symmetric setting, the budget ratio governs not only the shape family but also which probes are most useful for inverse recovery.

6 Structured Generalization Beyond One Knob

The next stage of the work tested whether the control law survives outside the exact symmetric ellipse case.

The first asymmetry pilot replaced the symmetric budget rule with a weighted split and found a clean structured upgrade:

- one-knob sufficiency fails under asymmetry,
- but a two-parameter family `(e, w)` collapses cleanly across scale,
- with maximum two-knob collapse error `3.7196e-08`.

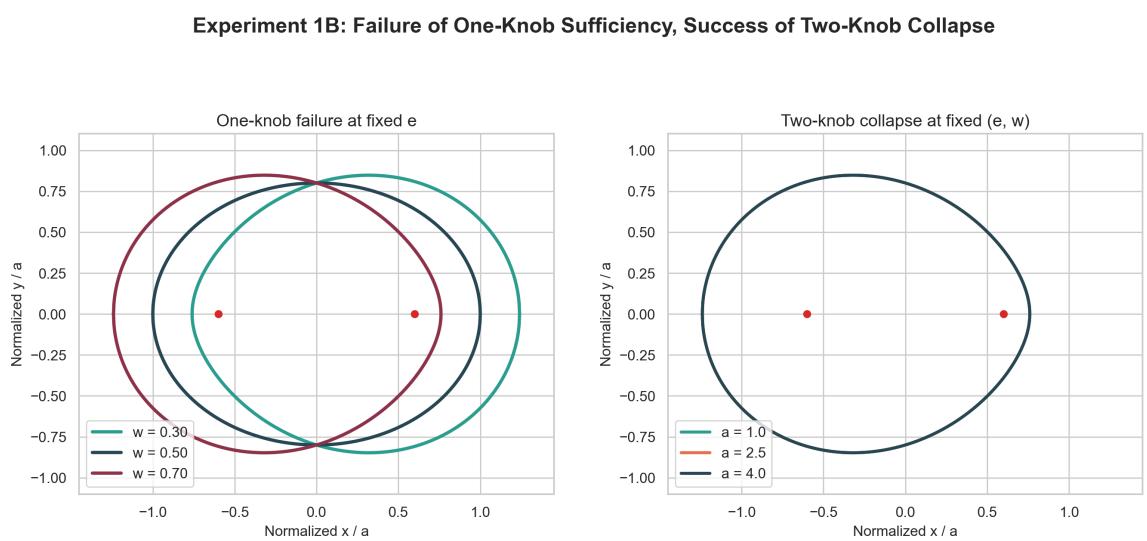

Figure 3: Asymmetry does not destroy the budget-governed structure; it upgrades it. Fixed `e` with varying weight `w` gives different normalized shapes, while fixed `(e, w)` still collapses across scale.

This is exactly the kind of result one wants if the principle is tracking real process structure at the family level.

The nearby extensions also stayed structured:

- the fixed-difference twin produced a hyperbola family with one-knob collapse under `lambda = a/c`,
- controlled quadratic anisotropy upgraded the raw family to `(e, alpha)` and whitening restored the original one-knob collapse,
- the equal-weight three-source case was organized by the normalized source triangle relative to budget and remained strongly low-dimensional,
- and the weighted three-source case was organized by the normalized source triangle plus the weight simplex, with the broad family behaving like a low-dimensional roughly five-parameter manifold.

The important point is that the principle did not survive as "still one scalar everywhere." It survived as a low-dimensional allocation geometry.

7 From Descriptor To Operational Latent Variable

The most important shift in the repository is inferential, not geometric.

In the weighted three-source canonical-pose inverse, a simple boundary-only reference-bank inverse recovered the normalized source triangle and normalized weights with useful accuracy:

- mean geometry MAE ranged from `0.062` to `0.086`,
- mean weight MAE ranged from `0.086` to `0.125`,
- and the weighted inverse beat an equal-weight baseline by about `1.88x` to `5.53x` depending on regime.

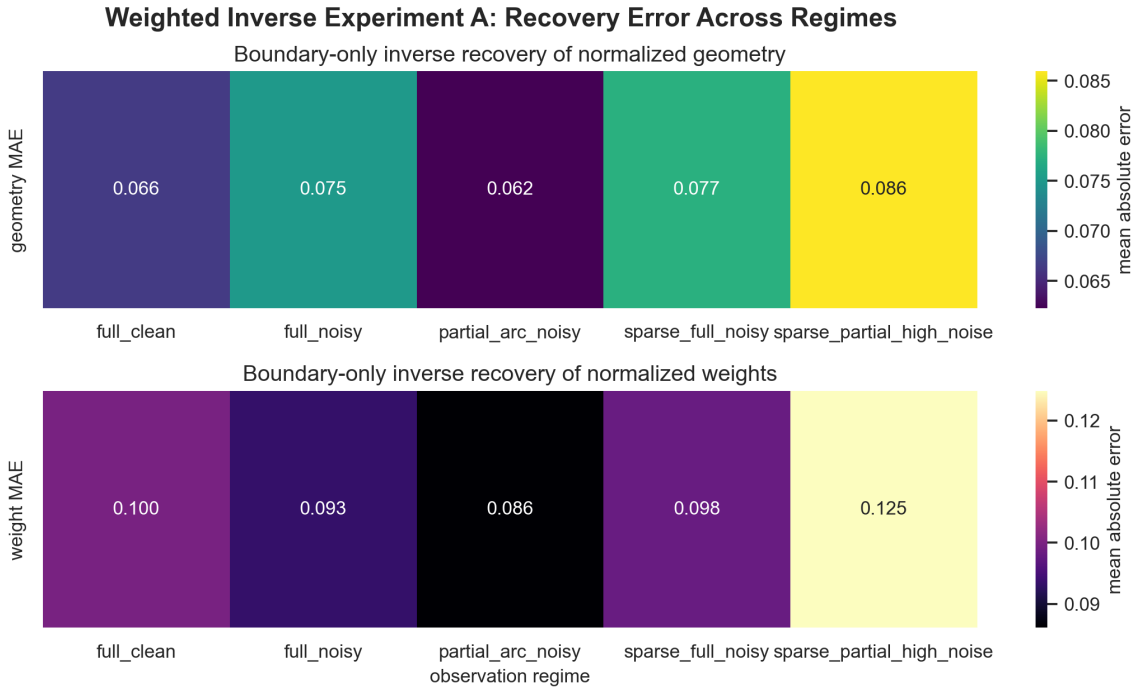

Figure 4: Weighted three-source inverse. The compact control object is not only a forward description of the family; it can be recovered from boundary data and clearly outperforms the equal-weight shortcut.

That is the point where the project stopped being only about how to describe shape and became about what hidden state can be inferred from it.

The controlled anisotropic extension strengthened that reading further. In the canonical-pose weighted anisotropic inverse:

- geometry MAE stayed around `0.077` to `0.089`,
- weight MAE stayed around `0.081` to `0.133`,
- mean `alpha` error stayed around `0.018` to `0.064`,
- and the anisotropy-aware inverse beat the Euclidean weighted shortcut by about `7.6x` to `14.0x`.

That is strong evidence that medium structure can join geometry and participation as part of the same operational latent object.

8 Pose-Free Anisotropy and the Current Solver Challenge

The hardest current branch is the pose-free anisotropic inverse, where unknown rotation and unknown medium anisotropy appear together.

In that combined-nuisance setting, the latent object remains operational but unevenly so:

- geometry MAE stayed around `0.071` to `0.104`,
- weight MAE stayed around `0.138` to `0.166`,
- but mean `alpha` error rose to about `0.146` to `0.304`.

This is not a uniform collapse of the inverse. It is a selective weakness along the anisotropy direction.

The matched ambiguity study made that diagnosis much sharper. Hiding rotation broadened the top-`10` `alpha` envelope by about `3.7x` to `5.6x` and worsened best-`alpha` error by about `11.4x` to `31.0x`, while geometry dispersion stayed essentially unchanged.

The oracle alignment ceiling then showed that the missing signal is largely still there. Giving the inverse the true pose improved `alpha` by about `5.65x` to `13.21x` across all regimes while leaving geometry roughly stable.

The newest failure-map result puts a shape on that solver challenge:

- full observations are the only region where simple practical locking stays broadly non-negative on oracle-gain capture,
- high-skew geometries are friendlier than low-skew and mid-skew ones,
- weak anisotropy is easier than moderate or strong anisotropy,
- and the main failure zones are sparse or partial observations with low-to-mid geometry skew.

The current pose-free anisotropy result is therefore not "BGP breaks." It is "the latent structure survives, but one latent direction is selectively masked by leftover symmetry before inference begins."

9 What The Repo Establishes

The repo now establishes the following results.

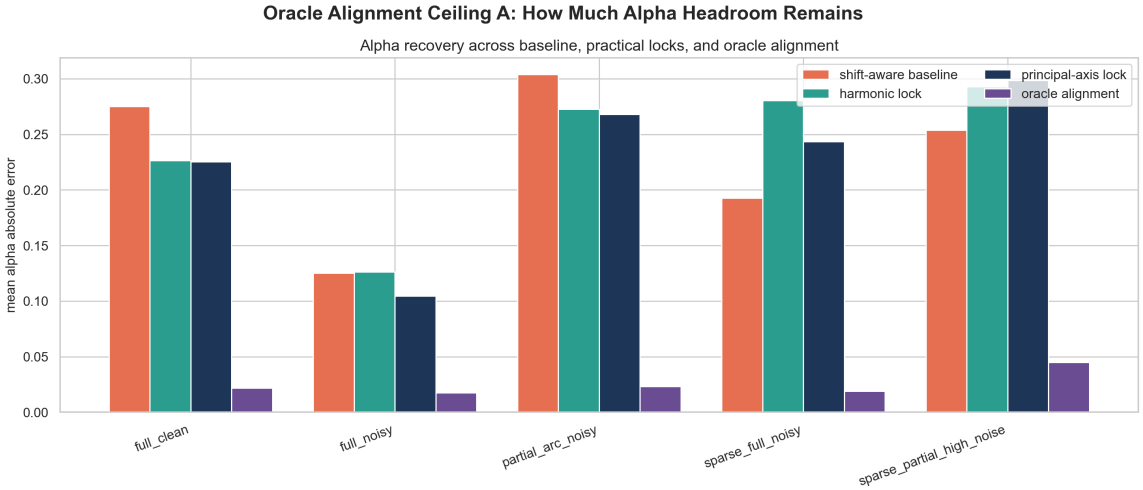

Figure 5: Oracle alignment ceiling. Once true pose is restored, most of the lost anisotropy signal comes back, which means the main solver challenge is practical symmetry handling rather than missing information in the boundary.

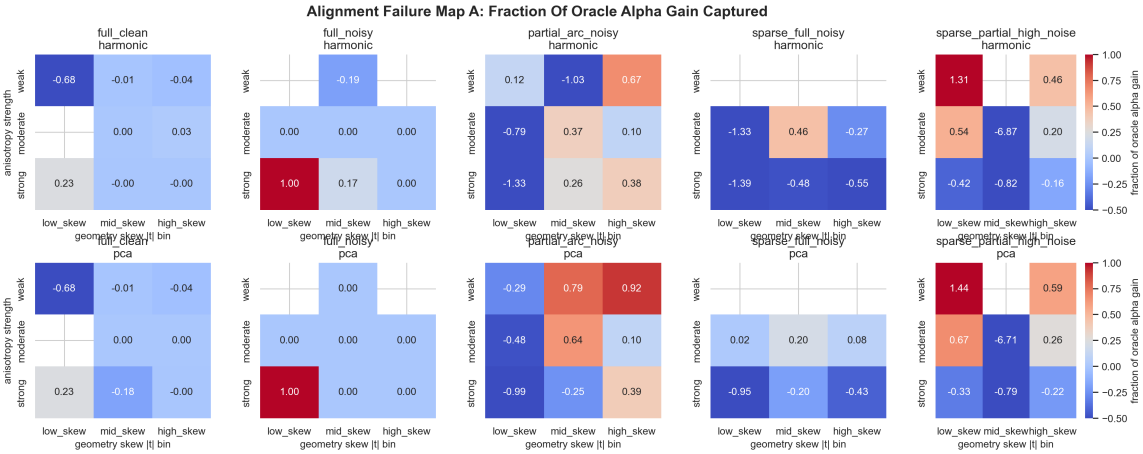

Figure 6: Alignment failure map. Practical pose handling fails in specific regions rather than uniformly, which means the remaining problem is now localized enough to target directly.

1. In the symmetric constant-sum two-source Euclidean process, `e = c/a` is the sufficient control variable for normalized geometry, and `b/a = sqrt(1-e^2)` is the corresponding transverse residue.
2. In that same symmetric known-source setting, `e` is operational: it is recoverable from noisy boundary observations and strongly outperforms raw separation and raw budget variables under scale shift.
3. Beyond the symmetric ellipse case, the budget-governor principle extends in a structured way. Asymmetry upgrades the family from one knob to two, controlled anisotropy adds a medium parameter, and three-source families are governed by compact normalized control objects rather than by one scalar.
4. In weighted multi-source settings, normalized geometry plus normalized participation forms an operational latent variable recoverable from boundary data.
5. In controlled anisotropic settings, medium structure can join that latent variable, and most of the current pose-free anisotropy penalty is a symmetry-handling problem rather than an absence-of-signal problem.

That is already a much larger result set than the original concept note.

10 Limits And Scope

This note is still narrower than a finished general theory.

These results are based on controlled computational studies. The strongest exact mathematical statement remains the symmetric two-source Euclidean core. Most of the richer claims are empirical structural claims established by numerical experiments rather than formal proofs.

The inverse results are also intentionally structured:

- finite reference banks rather than full continuous optimizers,
- radial-signature encodings rather than arbitrary boundary representations,
- controlled anisotropy rather than arbitrary warped media,
- and mostly three-source families rather than unrestricted source count.

The repository leaves these targets unresolved:

- universal sufficiency of one scalar beyond the symmetric ellipse case,
- recovery under arbitrary anisotropy-axis orientation,
- full robustness under richer non-quadratic or spatially varying media,
- or a final practical pose-equivariant inverse that closes the current anisotropy gap in sparse partial regimes.

Those are real unresolved technical targets, not wording details.

11 Practical Interpretation

The practical use of BGP is not “always compute eccentricity.”

The practical use is:

- identify the compact normalized budget variables that govern a family,
- use them as the primary state for prediction and comparison,

- and in inverse settings, ask whether the observed boundary is sufficient to recover that hidden budget object.

In the symmetric two-source case, that means `e` is a real control coordinate and even tells you which probes are worth trusting at different depletion phases.

In the richer multi-source and anisotropic families, the same logic becomes:

- normalized placement,
- normalized participation,
- and sometimes medium structure

are better state variables than raw distances, raw scale, or unstructured shape summaries.

The main practical engineering lesson from the current phase is just as important:

> when one part of the hidden budget state becomes weakly identifiable, the right next move is better symmetry handling rather than a larger bank or a more complicated regressor.

12 Next Revision Targets

The strongest next steps are:

1. build robust pose handling specifically for the sparse or partial low-to-mid-skew cells identified by the failure map,
2. run the planned probe-specialization experiment so the edge-regime conditioning results become a full inverse-design result rather than only a conditioning note,
3. test whether richer pose-equivariant or hybrid alignment methods can close a substantial fraction of the current oracle headroom,
4. extend the medium branch beyond a single axis-aligned anisotropy parameter,
5. and begin out-of-family application tests that use BGP as an inferential control object rather than only a geometric control law.

13 Conclusion

The narrow mathematical heart of the project is the symmetric two-source result: `e = c/a` is the budget governor for normalized geometry. The repository now establishes a broader result set as well. The same budget logic extends into low-dimensional control objects for asymmetric, anisotropic, and multi-source families, and in weighted inverse settings those objects form operational latent variables.

The current solver challenge is also clearer than it was at the start. The hardest remaining problem is not whether the latent state exists. It is how to keep enough symmetry broken before inference to recover the anisotropy part of that latent state robustly under sparse and partial observations.

That is a strong place for the program to be. BGP now functions as a technical framework for how budget-constrained geometry is organized and how much of that hidden organization can be recovered from what we observe.

14 Artifact References

- Concept note
- Derivation note
- Control-knob experiment
- Identifiability and baselines
- Asymmetry experiment
- Multi-source experiment
- Weighted multi-source inverse
- Weighted anisotropic inverse
- Pose-free weighted anisotropic inverse
- Latent ambiguity experiment
- Oracle alignment ceiling
- Alignment failure map
- Research roadmap

Lopez III, Dionisio Alberto. 2026. *Shape Budget*. Https://github.com/zfifteen/shape-budget.